

◦ Aim:

- To draw the I-V characteristics of a zenex diode and to determine the knee voltage (V_K), zenex voltage (V_Z), and dynamic resistance (R).

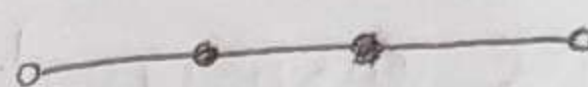
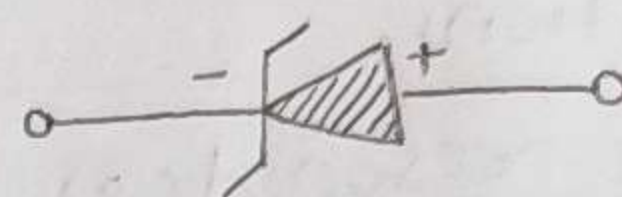
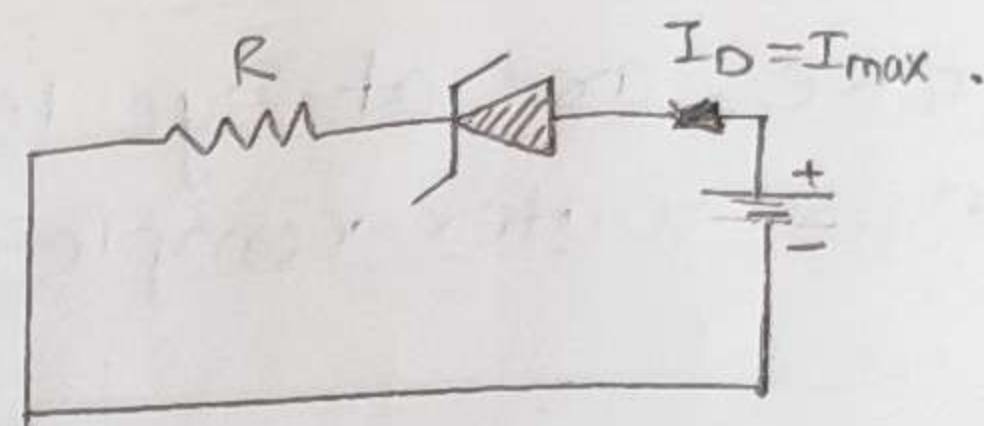
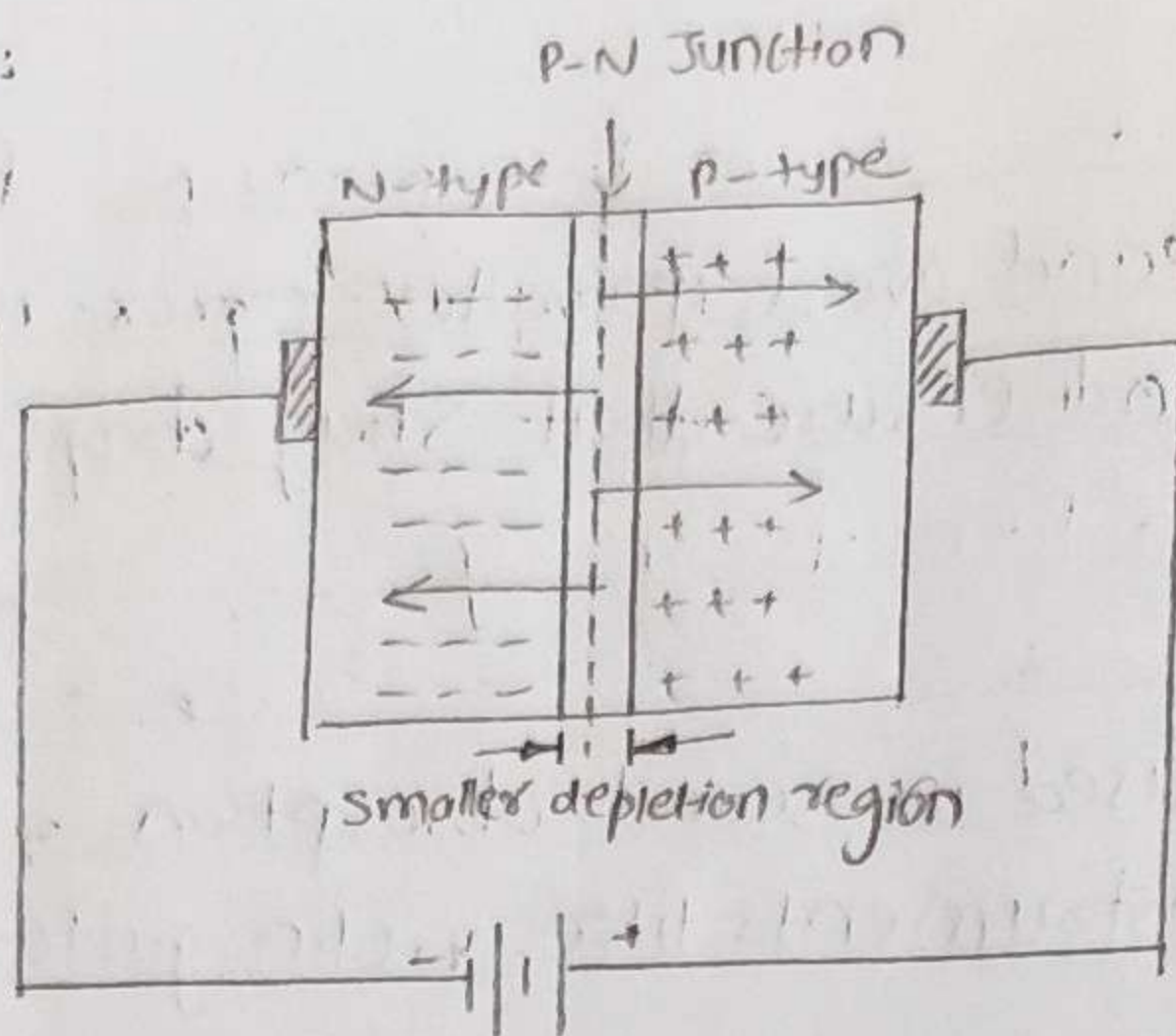
◦ Apparatus:

- zenex diode, DC power supply, Digital voltmeter, Digital Ammeter, Resistor control unit, connecting wires, etc.

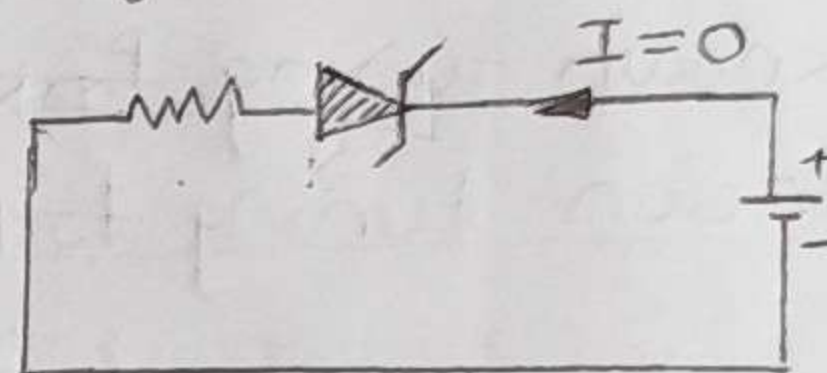
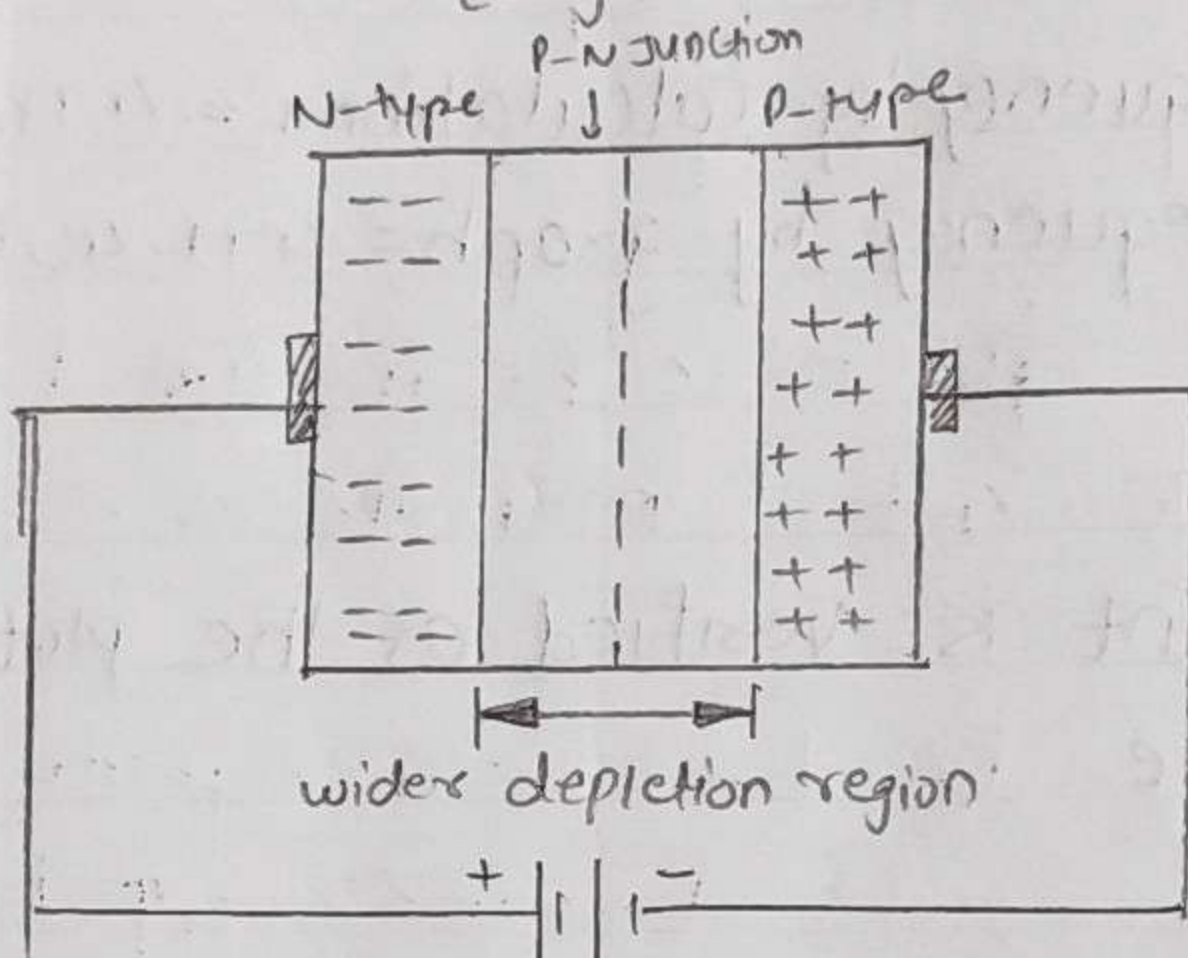
◦ Theory:

- Diode is a semiconductor device consisting of a junction between a n-type and a p-type semiconductor. The formation of this junction leads to a diffusion of a carrier across junction with the recombination of electrons and holes. As a result, the junction will be void of mobile charges but the transfer of charges results in a potential barrier.
- If an external bias is applied to the diode, the barrier potential may either increase or decrease on the direction of applied bias.
 - If p-type material is connected to the positive w.r.t. n-type material, it is called as 'forward bias'. Here, the potential barrier is lowered by applied bias and diode offers low resistance.
 - If n-type material is connected to the positive w.r.t. p-type material, it is called as 'reverse bias'. Here, the potential barrier is increased and the diode offers a higher resistance.

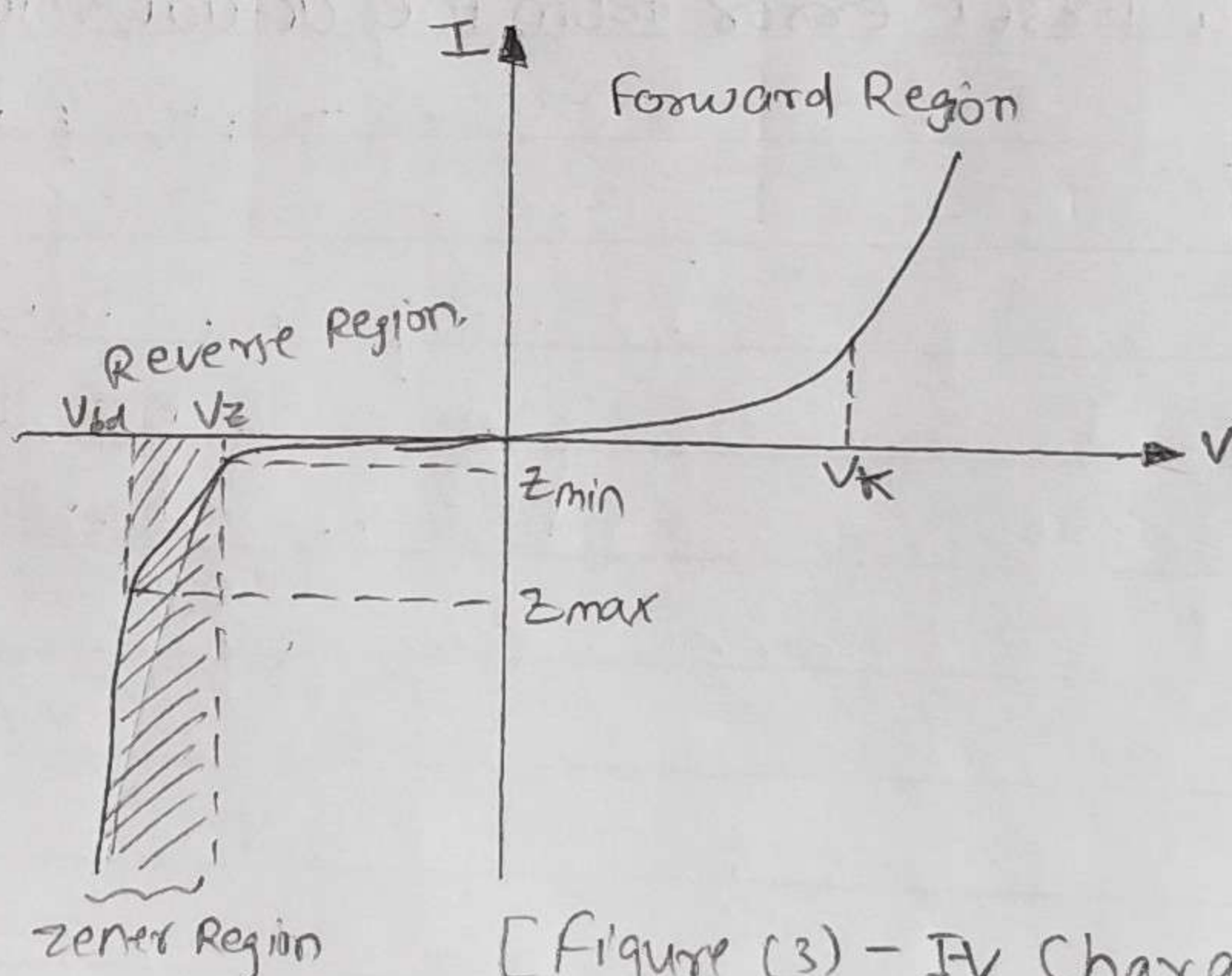
Diagrams:



[Figure (1) - Forward Biasing]



[Figure (2) - Reverse Biasing]



[Figure (3) - IV Characteristics]

Observation Table:

Sr. No.	Forward Voltage	Forward Current	Reverse Voltage	Reverse Current
1	0	0	0	0
2	0.13	0	0.87	0
3	0.25	0	2.12	0
4	0.35	0	3.62	0
5	0.53	0	6.97	0
6	0.60	0	11.43	0
7	0.70	0.70	→ 11.76 (V_Z)	0.10
8	0.73	1.90	11.82	0.70
9	0.74	2.70	11.87	1.70
10	0.75	3.60	→ 12.00	4.60
11	0.76	4.40	12.06	6.50
12	0.77	5.40	12.19	9.30
13	0.78 (V_K)	7.40	12.27	12.80
14	0.79	8.90	12.31	13.50
15	0.80	12.60	12.37	16.30
16	0.81	15.80	12.43	18.00
17	0.82	18.20	12.47	18.60
18	0.83	24.50	12.50	19.90
19	0.84	26.50	12.53	20.70
20	0.85	30.20	12.57	21.10

Calculations: (from graph)

$$R = \frac{\Delta V}{\Delta I} = \frac{(12.00 - 11.76)V}{(4.60 - 0.11)mA} = \frac{0.24}{4.5} k\Omega = 0.0533 k\Omega$$

Hence, $R = 53.33 \Omega$.

- Current across the junction is given by:

$$I = I_0 [\exp(eV/kT) - 1]$$

where I_0 is the saturation current

e is the electron charge $= 1.6 \times 10^{-19}$

V is the applied voltage

k is the Boltzmann's constant $= 1.38 \times 10^{-23} \text{ J/K}$

T is the Room Temperature

- A Zener diode is a diode which has its p-type and n-type materials heavily doped and hence its depletion width is narrow. As a result its depletion reverse bias voltage is much high before sharply breaking.

° Procedure:

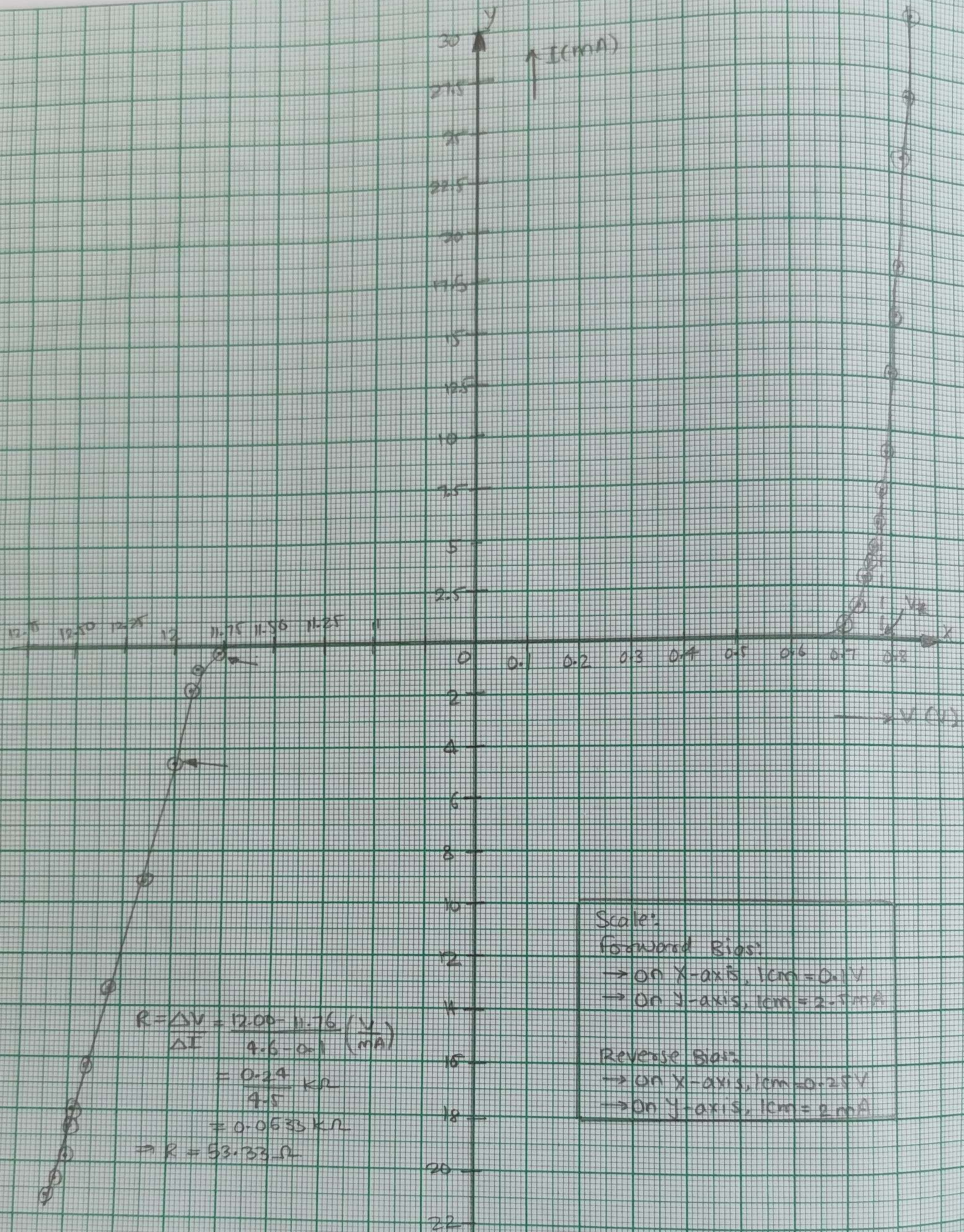
(i) Forward Biasing:

- Connect positive side to p-type material and negative side to n-type material. Increase the voltage slightly slowly until you get a sharp increase in the current. Take a note of the voltages and currents. The above voltage is your 'knee voltage (V_k)'.

- Refer Figure (1).

(ii) Reverse Biasing:

- Connect positive side to n-type material and negative side to p-type material. Increase reverse voltage slowly until you get some current. Note the voltages and currents as you slowly increase the voltage. Once it starts increasing, the corresponding voltage is 'Zener voltage (V_z)', and at point where current increases significantly is the 'Breakdown Voltage (V_{bd})'. The region between V_z and V_{bd} is called 'Zener Region'.



- Take 2 points in the zener region after plotting your observations on a graph of current (I) vs. voltage (V). The slope between the straight line connecting the two points is the 'dynamic resistance (R)'.
- Refer Figures (2) and (3).

- o Precautions:

- Make sure that circuit is connected properly and no short circuit.
- Do not touch live wires.

- o Applications:

- zener diode is used as voltage regulators.

- o Result:

- Knee voltage (V_k) = 0.78V
- Zener voltage (V_z) = 11.76V (rev)
- Dynamic Resistance (R) = 53.33 Ω .

- o Conclusion:

- The I-V characteristics of zener diode is plotted.
- The knee voltage, zener voltage, and dynamic resistance are found out to be $V_k = 0.78V$, $V_z = 11.76V$, and $R = 53.33 \Omega$ respectively.